

AI Lead Capture & Auto Response System (n8n Workflow Blueprint)

Overview

This workflow automates the entire lead response process using AI.

It captures incoming leads, generates intelligent responses, logs all interactions, and sends instant email replies — ensuring no lead is lost due to delayed communication.

Problem Statement

Businesses often lose potential customers because:

- Responses are delayed
 - Leads are not tracked properly
 - No system exists to monitor communication
-

Solution

This workflow provides:

- ⚡ Instant AI-generated responses
 - 📊 Complete lead and response tracking
 - 📧 Automated email communication
 - 🔄 Status monitoring (pending/sent/failed)
-

System Architecture

Webhook → Data Processing → AI Response → Merge Data → Logging → Email → Status Update

Workflow Breakdown (Node-by-Node)

1 Webhook (Trigger)

- Captures incoming lead data
- Accepts:
- Name

- Email
 - Message
-

2 Edit Fields (Data Cleaning)

- Standardizes incoming data
 - Ensures consistent structure
-

3 Code Node (Generate Lead ID)

- Generates unique lead ID
 - Adds timestamp
-

4 Google Sheets (Leads Storage)

- Stores raw lead data
 - Acts as primary database
-

5 AI Agent (OpenAI)

- Generates contextual response
 - Input:
 - User message
 - Output:
 - Personalized reply
-

6 Merge Node (Data + AI Output)

- Combines:
 - Lead data
 - AI response
-

7 Code Node (Response Structuring)

Creates structured response object:

```
{
  "response_id": "...",
  "lead_id": "...",
  "email": "...",
  "ai_response": "...",
}
```

```
"status": "pending"
}
```

8 Google Sheets (Response Logging)

- Stores AI responses
- Maintains audit trail

9 Gmail Node (Send Email)

- Sends AI-generated reply to user
- Uses dynamic fields

10 Merge Node (Preserve Data)

- Combines Gmail response with original data
- Prevents data loss

1 1 Google Sheets (Update Status)

- Updates status:
- pending → sent
- pending → failed (if error handling added)

Data Flow

Stage	Output
Lead Capture	Raw user data
AI Processing	Generated response
Logging	Stored records
Delivery	Email sent
Tracking	Status updated

Error Handling (Recommended)

Add IF Node after Gmail:

- If success → status = sent

- If failure → status = failed
-

Setup Instructions

1. Import workflow JSON into n8n
 2. Configure credentials:
 3. OpenAI API Key
 4. Google Sheets
 5. Gmail
 6. Update Sheet columns:
 7. Leads Sheet
 8. Responses Sheet
 9. Activate workflow
-

Required Integrations

- OpenAI API
 - Google Sheets
 - Gmail
-

Use Cases

- Lead generation systems
 - Marketing funnels
 - Customer support automation
 - Freelancers & agencies
-

Benefits

- Instant response time
 - Improved lead conversion
 - Full transparency
 - Scalable automation
-

Future Enhancements

- WhatsApp integration

- CRM integration
 - Lead scoring (AI)
 - Dashboard analytics
-

Author

Built as part of a real-world AI automation series.

Conclusion

This workflow demonstrates how AI and automation can be combined to solve real business problems efficiently and reliably.